**Project Design Phase-II**
**Solution Requirements (Functional & Non-functional)**

| | |
|---|---|
| Date | 28 October 2026 |
| Team ID | EN25CS3010384 |
| Project Name | Electric Vehicle Market Analysis Using Tableau |
| Maximum Marks | 4 Marks |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Data Access & Integration | The solution should load the EV dataset containing vehicle categories, sales, manufacturers, manufacturing countries, battery types, driving range, charging capacity, efficiency, and environmental metrics. | High |
| 2 | Data Processing | The solution should clean and organize the dataset, handle missing values where necessary, and prepare data for analysis in Tableau. | High |
| 3 | Business Analysis | The solution should calculate and compare annual sales, average driving range, sales trends, manufacturer performance, battery technology distribution, charging efficiency, and regional environmental impact. | High |
| 4 | Business Insights & Analytical Outputs | The solution should provide nine visualizations across three scenarios: Emily's market analysis, David's manufacturer analysis, and Sarah's sustainability analysis. | |
| 5 | User Interaction & Analysis | Users should be able to explore charts, view tooltips, compare categories and manufacturers, and analyze trends using Tableau dashboard interactions. | Medium |
| 6 | Data Update & Reporting | The solution should allow the dataset to be refreshed when updated data is available and enable users to export or share dashboards and analytical findings. | Medium |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description |
|---|---|---|

| | | |
|---|---|---|
| NFR-1 | **Usability** | Dashboards should have clear titles, readable labels, appropriate colors, and an easy-to-understand layout. |
| NFR-2 | **Security** | Project files and dataset access should follow applicable access permissions, and shared links should be configured appropriately. |
| NFR-3 | **Reliability** | Visualizations should display consistent results based on the source dataset and correctly configured calculations. |
| NFR-4 | **Performance** | Dashboards should load and respond efficiently, with optimized worksheets and calculations. |
| NFR-5 | **Availability** | The published Tableau dashboard should be accessible through its sharing link when the hosting service is available. |
| NFR-6 | **Scalability** | The workbook should support additional vehicle records and updated data without requiring the entire dashboard to be redesigned. |